Combinatorial Algorithms (Math 482), 2026 Spring   Problem Set 10

Due by Wednesday, April 8, 2026.

1. **Checking Correctness.** We are given a directed graph $G = (V, E)$ with n vertices $V = \{v_1, \dots, v_n\}$, m edges, and a source vertex s. We are also given an array $D[1..n]$. Design an algorithm that checks whether $D[i]$ equals the length of the shortest path from s to v_i for all i. Your algorithm should run in $O(m + n)$ time. (This means that we cannot afford to run Dijkstra's algorithm: that would take $O(m \log n)$ time.)

2. **Transitive closure.** For a directed acyclic graph $G = (V, E)$, the *transitive closure* is a graph $H = (V, F)$ constructed as follows: If G contains a directed path from u to v, then add a directed edge (u, v) to H. If G has n vertices and m edges, design an $O(mn)$-time algorithm that computes the transitive closure of G.

3. **Fault-tolerant paths.** We are given a directed graph $G = (V, E)$ with vertices $s, t \in V$. We would like to find two paths from s to t such that the two paths do not share any edges (the paths are *edge-disjoint*). Show how to reduce this problem to a network flow problem. Design an algorithm that either finds two edge-disjoint paths or reports that no such paths exist. What is the running time of your algorithm?

   (a) Does your reduction generalize to finding k edge-disjoint paths from s to t?

   (b) How does the running time depend on k?

4. **Vertex capacities.** We are given a capacitated network (a directed graph G with edge capacities $c(e) \geq 0$, and an s-t pair). However, we are also given additional constraints: Each vertex v has a given capacity $c(v)$, which means that the total flow entering (or leaving) vertex v cannot exceed $c(v)$. We would like to maximize the flow from s to t subject to these constraints.

   Design an algorithm for this problem, by reducing it to a network flow problem. Prove the correctness of your algorithm.

5. **Convex combination of flows.** In a capacitated network, we label the edges from 1 to m as $e_1, e_2, \dots, e_m$. We represent a flow f as an array $A[1..m]$, where $A[i] = f(e_i)$ is the flow on edge i. Suppose that we are given two flows, f_1 and f_2, represented by the arrays $A[1..m]$ and $B[1..m]$.

   A convex combination of $A[1..m]$ and $B[1..m]$ is an array $C[1..m]$ such that $C[i] = c \cdot A[i] + (1 - c) \cdot B[i]$ for a constant c, $0 \leq c \leq 1$. Show that the convex combination of $A[1..m]$ and $B[1..m]$ is also a flow (that is, $C[1..m]$ satisfies all flow conservation and the capacity constraints).